



ALLIANCE UNIVERSITY

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Mini-Project Report

Bachelor of Computer Applications

Semester IV

Full Stack Development

By

Vishesh Gurnani

2411021240033

Department of Computer Applications

Alliance University

Chandapura Anekal Main Road, Anekal

Bengaluru 562106

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Introduction

What is MongoDB?

MongoDB is a popular, open-source NoSQL database designed for high performance, flexibility, and scalability, storing data in flexible, JSON-like documents instead of traditional tables. It is ideal for modern, fast-moving applications, supporting rapid development, horizontal scaling, and varied data structures, with MongoDB Atlas offering a cloud-native, fully managed version.



What is the MongoDB Community Server?

MongoDB Community Server is the free, open-source edition of MongoDB that allows developers to run a NoSQL document database locally on their own system. It stores data in flexible, JSON-like documents within collections, making it well-suited for modern web applications that require dynamic schemas. In the context of CRUD (Create, Read, Update, Delete) operations, MongoDB Community Server enables users to create new documents, retrieve existing data, modify records, and delete documents efficiently using its query language, making it a practical tool for learning and implementing database management concepts in backend development.

What is the MongoDB Compass?

MongoDB Compass is the official graphical user interface (GUI) for MongoDB that allows users to interact with their database visually instead of using command-line queries. It provides an easy way to perform CRUD (Create, Read, Update, Delete) operations by allowing users to insert documents, view and filter data, modify existing records, and delete documents through an intuitive interface. Compass also includes features such as schema visualization, performance insights, and index management, making it especially useful for beginners and developers who want to explore and manage MongoDB databases efficiently without writing complex commands.

Difference between MongoDB Community Server and Compass

<i>Feature</i>	<i>MongoDB Community Server</i>	<i>MongoDB Compass</i>
Type of Tool	Database server (backend engine)	Graphical User Interface (GUI) application
Primary Purpose	Stores and manages data	Allows users to visually interact with MongoDB databases
Role in CRUD	Executes CRUD operations at the database level	Provides a visual way to perform CRUD operations
Interface	Command-line based (mongosh) or accessed via drivers	Visual interface with buttons, forms, and filters
Installation	Runs as a background service on a system	Installed as a desktop application
Data Storage	Actually stores collections and documents	Does not store data; connects to an existing MongoDB server
Technical Requirement	Requires knowledge of queries and commands	Beginner-friendly; minimal command knowledge needed
Use Case	Backend development and application deployment	Database exploration, testing, and management
Dependency	Can run independently	Requires MongoDB Community Server or Atlas to connect

Installation

1. Download the latest MongoDB 4.4.x version from the Download MongoDB Community Server page. (<https://www.mongodb.com/try/download/community>)
2. In the Available Download section, select the version to download.
 - a. From the Platform field, select Windows.
 - b. From the Package field, select msi.
You have the option to download a msi or zip file.
 - c. Select Download.
3. Navigate to the folder where the MongoDB installation file was downloaded and run

the msi file.

This starts the installation process.

4. Follow the MongoDB Setup Wizard to complete the install.

Select Next to view the license agreement. To continue, accept the license and select Next.

5. On the Choose Setup Type window, select Complete.

6. On the Service Configuration window, specify the following options and then select Next.

- Check Install MongoDB as a Service. It is required to run MongoDB as a Windows service.
- Enable Run service as Network Service user.
- You can use the default values for Service, Data Directory, and Log Directory. If you modify these values, be sure to keep a note of the values.

They are needed for the configuration steps.

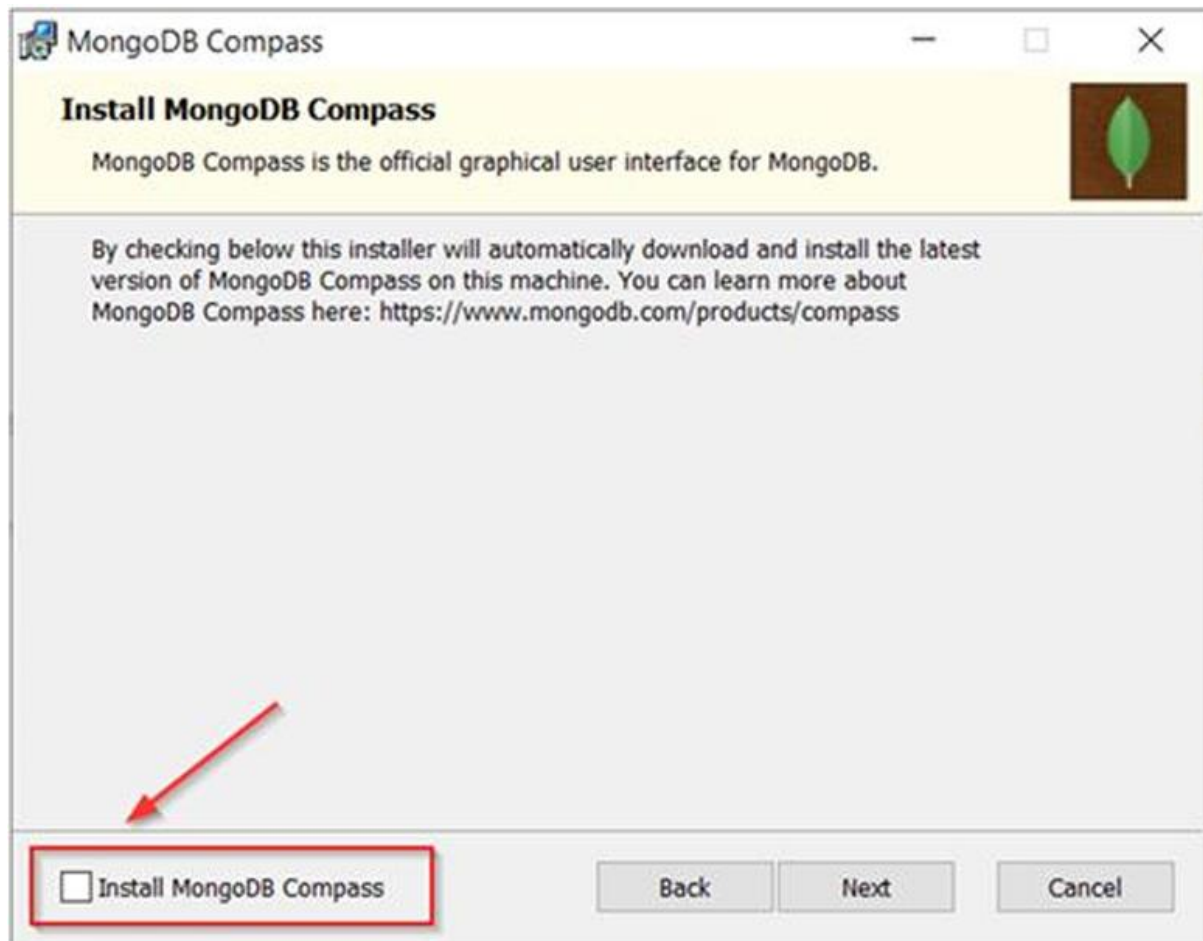
The screenshot shows the 'Service Configuration' window of the MongoDB Setup Wizard. The title bar is 'Service Configuration' and the subtitle is 'Specify optional settings to configure MongoDB as a service.' The window contains several configuration options:

- ☒ **Install MongoDB as a Service**
- ☒ **Run service as Network Service user**
- ☐ **Run service as a local or domain user:**
 - Account Domain: [.]
 - Account Name: [MongoDB]
 - Account Password: []
- Service Name: [MongoDB]
- Data Directory: [C:\Program Files\MongoDB\Server\4.4\data\]
- Log Directory: [C:\Program Files\MongoDB\Server\4.4\log\]

At the bottom right, there are three buttons: '< Back', 'Next >', and 'Cancel'.

7. Install MongoDB Compass is optional and not required for deploying Sametime.

Clear the checkbox for Install MongoDB Compass and select Next.



(If MongoDB Compass is installed, it starts after the installation process completes. You can shut down the MongoDB Compass application, as it is not used for Sametime deployments.)

8. Select Install to complete the MongoDB installation.

9. When the installation process is complete, select Finish.

10. After the system restarts, MongoDB automatically runs as a Windows service.



CRUD Operations

CRUD stands for:

C → **Create**

R → **Read**

U → **Update**

D → **Delete**

1. Create (Inserting Documents):

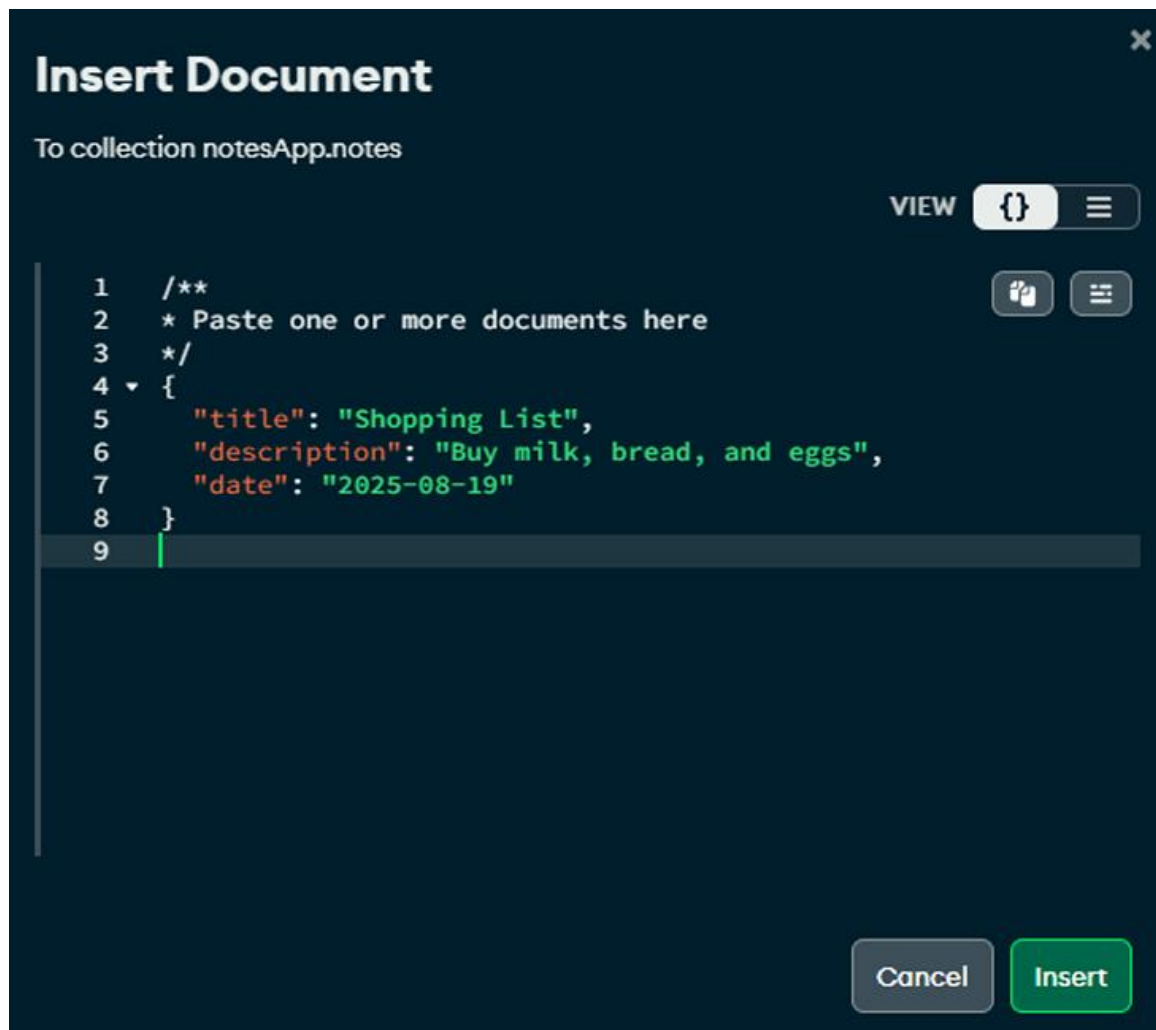
When we want to add new data into our database, we use Insert.

Example: Inserting Notes

Steps in Compass:

- 1) Open Compass → connect to `mongodb://localhost:27017`
- 2) Create a Database → name: `notesApp`
- 3) Create a Collection → name: `notes`
- 4) Click on Insert Document
- 5) Add this JSON:

```
{  
  "title": "Shopping List",  
  "description": "Buy milk, bread, and eggs",  
  "date": "2025-08-19"  
}
```



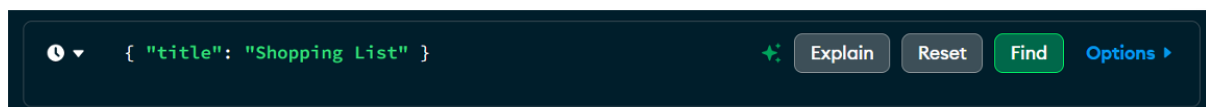
2. Read (Find Documents)

Reading data is the most common operation. In Compass, this is done using the Filter bar.

Example: Find Notes by Title

1. Go to the notes collection.
2. In the Filter bar, type:

{ "title": "Shopping List" }



3. Update (Modify Documents)

Sometimes you need to update data, like correcting spelling or changing values.

Example: Updating a Note's Description

1. Select the note with title "Shopping List".
2. Click on Edit Document.
3. Change description from:
"Buy milk, bread, and eggs"
to
"Buy milk, bread, eggs, and butter".
4. Click Update

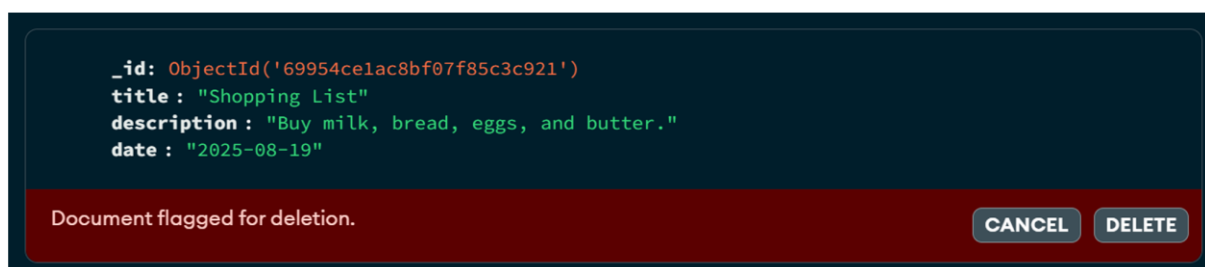


4. Delete (Remove Documents)

If you no longer need some data, you can delete it.

Example: Delete One Note

- 1) In the notes collection, click on the note with "title": "Shopping List".
- 2) Click on Delete Document.
- 3) Confirm → Note deleted.



Conclusion

This mini project on CRUD Operations in MongoDB helps to grasp how data is managed with MongoDB Community Server and MongoDB Compass. It covered the essential ideas behind MongoDB as a NoSQL, document-oriented database and showed how it stands apart from traditional relational databases. The report clearly lays out the installation and configuration process, making it easy to understand how to set up MongoDB as a Windows service. By diving into Create, Read, Update, and Delete operations with MongoDB Compass, the project illustrated how efficiently data can be inserted, retrieved, modified, and deleted within a collection. The step-by-step examples, particularly those managing records in the notesApp database, made it straightforward to grasp real-time database interaction in a clear and organized way.